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Windows PowerShell
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D4_LlamaIndex_Ollama> python w7d4_ollama_local_rag.py
=====
W7D4 - LLAMAINDEX + OLLAMA - LOCAL RAG
=====

Ollama API:
http://localhost:11434/api/chat

Models:
- llama3.2:3b
- qwen2.5:3b

=====
TASK 1 - FIRST LOCAL INFERENCE
=====

Prompt:
Explain what a local LLM is in two sentences.

Response:
A local Large Language Model (LLM) is a type of machine learning model that is specifically designed to operate on a local device, such as a computer or server, rather than in a cloud-based environment. This means that a local LLM stores its weights and models locally, allowing for faster inference and lower latency, but also requiring more computational resources and storage space to train and maintain.

Response time: 16.65 seconds

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TASK 2 - FIVE PROMPTS WITH CUSTOM SYSTEM PROMPT
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Prompt 1
-----
What is an LLM?

Response:
**What is an LLM?**
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Response:
**What is an LLM?**

An LLM, or Large Language Model, is a type of artificial intelligence (AI) designed to understand and generate human-like language. It's a computer program that's trained on massive amounts of text data to learn patterns and relationships in language.

**How is it trained?**

LLMs are trained using a process called self-supervised learning, where the model learns to predict the next word in a sentence or sequence based on the context of the previous words. This is done using a massive corpus of text data, which the model uses to learn patterns and relationships in language.

**What are the capabilities of an LLM?**

LLMs can perform a variety of tasks, such as:

* **Text generation**: generating human-like text based on a prompt or topic
* **Language translation**: translating text from one language to another
* **Question answering**: answering questions based on a given text or context
* **Text summarization**: summarizing long pieces of text into shorter ones

**What's the goal of an LLM?**

The goal of an LLM is to create a model that can understand and generate human-like language, which has many applications in areas such as:

* **Chatbots**: enabling computers to have conversations with humans
* **Content generation**: generating articles, social media posts, and other types of content
* **Language translation**: enabling computers to translate languages in real-time

I hope this helps! Do you have any specific questions about LLMs or their applications?

Response time: 46.87 seconds

Prompt 2
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What is RAG and why is it useful?
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Response:
RAG stands for "Reinforced Attention Guidance." It's a technique used in reinforcement learning (RL) to help agents focus on specific tasks or goals.

In RL, an agent learns to take actions to maximize a reward signal. However, the agent might get distracted by other aspects of the environment, leading to suboptimal performance. RAG is designed to help the agent stay focused on the task at hand.

Here's how it works:

1. The agent learns to recognize specific cues or features in the environment that indicate the task is relevant (e.g., a specific object or a target location).
2. When the agent detects these cues, it triggers a "reinforced attention" mechanism, which increases the agent's motivation to focus on the task.
3. The agent then takes actions that maximize the reward signal, while maintaining attention on the relevant cues.

RAG is useful for several reasons:

* Improved task performance: By keeping the agent focused on the task, RAG can lead to better performance on complex tasks that require attention and concentration.
* Reduced exploration: RAG can help reduce the agent's exploration of the environment, which can be beneficial in situations where exploration is costly or unnecessary.
* Increased robustness: RAG can make the agent more robust to changes in the environment or noise, as the agent is more likely to stay focused on the task despite distractions.

RAG has been successfully applied in various RL tasks, including robotic control, game playing, and autonomous driving.

Response time: 45.12 seconds

Prompt 3
-----
Explain vector embeddings in simple terms.

Response:
Vector Embeddings: A Simple Explanation

Imagine you're trying to describe a cat. You could use words like "furry", "small", "cute", and so on. But, what if you could repre
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nt a cat as a single, compact number (or vector) that captures its essence? That's basically what vector embeddings do!

What are Vector Embeddings?

Vector embeddings are a way to represent words, objects, or concepts as numerical vectors in a high-dimensional space. These vectors capture the meaning or characteristics of the original entity, allowing us to perform operations like similarity searches, clustering, and classification.

Think of it like a Map

Imagine a map of the world. Each country is represented by a unique point on the map. Now, imagine you want to find all countries that share a similar geography, culture, or climate. You wouldn't look at each country separately; instead, you'd look at the map as a whole, identifying patterns and similarities.

Similarly, vector embeddings create a "map" in the form of a high-dimensional space, where similar words or concepts are mapped to nearby points. This allows us to identify patterns and relationships between entities.

Example: Word Vectors

For example, consider two words: "dog" and "cat". If we were to represent them as vector embeddings, the resulting vectors might look something like this:

* Dog: [1, 0, 0, 0, 1] (high on the first axis, low on the others)
* Cat: [0, 1, 0, 0, 1] (high on the second axis, low on the others)

These vectors capture the essence of each word, highlighting the similarities (or differences) between them.

What's the Benefit?

Vector embeddings have many applications, such as:

* Text analysis: finding related words or documents
* Image recognition: identifying objects or patterns
* Recommendation systems: suggesting products based on user preferences

The key advantage is that vector embeddings allow us to perform complex operations on high-dimensional data using simple, intuitive algorithms.
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I hope this explanation helped you understand vector embeddings! Do you have any specific questions or topics you'd like me to elaborate on?

Response time: 69.17 seconds

Prompt 4
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What is the difference between AI and machine learning?

Response:
AI (Artificial Intelligence) and Machine Learning (ML) are often used interchangeably, but they are not exactly the same thing. Here's a simple explanation:

Artificial Intelligence (AI):

AI refers to the broader field of research and development aimed at creating intelligent machines that can perform tasks that typically require human intelligence, such as:

* Reasoning and problem-solving
* Perception and understanding
* Learning and adaptation
* Language understanding and generation
* Expertise and decision-making

AI encompasses a wide range of techniques, including rule-based systems, expert systems, and machine learning. The goal of AI is to create machines that can think, learn, and behave like humans.

Machine Learning (ML):

Machine Learning is a subset of AI that focuses specifically on developing algorithms and statistical models that enable machines to learn from data, without being explicitly programmed. In other words, ML is a way to teach machines to improve their performance on a task by learning from experience.

ML involves training algorithms on large datasets to enable them to recognize patterns, make predictions, and make decisions. The goal of ML is to create machines that can automatically improve their performance on a task, without requiring explicit programming.
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Key differences:

1. Scope: AI is a broader field that encompasses multiple techniques, including ML. ML is a specific subset of AI.
2. Focus: AI focuses on creating intelligent machines that can perform tasks that require human intelligence. ML focuses on developing algorithms that enable machines to learn from data.
3. Goals: AI aims to create machines that can think, learn, and behave like humans. ML aims to create machines that can automatically improve their performance on a task by learning from data.

To illustrate the difference, consider a self-driving car. AI is the broader field that enables the car to recognize objects, understand language, and make decisions. ML is the specific technique used to train the car's sensors to recognize objects, such as pedestrians or traffic lights, based on large datasets of images and sensor data.

I hope this helps clarify the difference between AI and Machine Learning!

Response time: 63.30 seconds

Prompt 5
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Why can a local LLM be useful for privacy?

Response:
A local LLM (Large Language Model) can be useful for privacy because it doesn't rely on a cloud-based server that can access and store user data. Here's why:

1. Data storage: When using a cloud-based LLM, your data is stored on the server, which can be accessed by others, including the company operating the server, law enforcement, or even hackers. A local LLM, on the other hand, stores your data on your own device, keeping it private and under your control.
2. Data transmission: When you interact with a cloud-based LLM, your data is transmitted to the server, which can be intercepted by third parties. A local LLM doesn't require data transmission to the cloud, reducing the risk of eavesdropping.
3. Model updates: Cloud-based LLMs often receive model updates from the company, which can contain sensitive information. A local LLM can be updated locally, without relying on cloud-based updates.
4. Data protection: Local LLMs are bound by the data protection policies of your device, such as GDPR or CCPA, which can provide stronger privacy guarantees than cloud-based LLMs.

However, it's essential to note that a local LLM is not completely "private" and may still be vulnerable to certain threats, such as:
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* Side-channel attacks (e.g., timing or spectral attacks)
* Inference attacks (e.g., using other data to infer information about the local model)
* Physical attacks (e.g., extracting data from the local device)

To mitigate these risks, it's crucial to use a local LLM with proper security measures, such as:

* Using a reputable and secure LLM model
* Implementing additional security measures, like encryption and secure boot
* Regularly updating the local LLM to ensure it has the latest security patches

By using a local LLM and taking necessary security measures, you can enjoy the benefits of private data storage and processing while minimizing the risks.

Response time: 62.21 seconds

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TASK 3 - LLAMA 3.2 vs QWEN 2.5 COMPARISON
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Question 1
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Explain Retrieval-Augmented Generation (RAG) in simple terms.

LLAMA 3.2:3B
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Retrieval-Augmented Generation (RAG) is a technique used in Natural Language Processing (NLP) to improve the performance of text generation models.

**What does it do?**

Imagine you're trying to write a short story. You start by brainstorming ideas, but then you need to recall specific details, such as characters' names, plot twists, or settings. RAG helps to retrieve this information from a database or a knowledge base, and then uses it to generate the next part of the story.

**How does it work?**
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A RAG system consists of two main components:

1. **Retriever**: This is a search engine that looks up the required information in the knowledge base.
2. **Generator**: This is the text generation model that uses the retrieved information to generate the next part of the story.

Here's a simple example:

* The retriever searches for information about a specific character in a knowledge base.
* The generator uses the retrieved information to write a paragraph about that character.

**Benefits**

RAG improves the performance of text generation models in several ways:

* **Increased accuracy**: By retrieving relevant information, the generator can generate more accurate and coherent text.
* **Improved completion**: RAG can help generate text that is more likely to be completed, by providing the necessary information to move the story forward.

In summary, RAG is a technique that combines retrieval and generation to improve the performance of text generation models. It helps to retrieve relevant information from a knowledge base and use it to generate more accurate and coherent text.
Response time: 45.76 seconds

QWEN 2.5:3B
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Sure, let's think of it in simple terms:

Imagine you have a big library (this is like your knowledge base) and you want to find a specific book (this is like finding the right answer to a question). In the old days, you would search through every shelf (this is like a traditional search engine) to find the book.

But now, there's a new method called Retrieval-Augmented Generation (RAG). It's like having a super helpful librarian who knows where all the books are. First, you use the old search method to find the librarian (this is like using a traditional search engine). Then, you ask the librarian (this is like asking a model or a tool that understands text) where the book might be.

This way, you can get more precise answers because the librarian knows exactly where to look for the book, and the model can understand the question and give you a better answer. It's like having a very smart and knowledgeable librarian who can not only find books b
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nd the question and give you a better answer. It's like having a very smart and knowledgeable librarian who can not only find books b
ut also understand what you're asking for and give you the exact answer you're looking for.

So, RAG is a method that combines the power of search (finding the right person to ask) with the power of generation (getting a bette
r answer to your question).
Response time: 72.33 seconds

Comparison:
Both models were tested with the same question and the same custom system prompt.

Question 2
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What is the difference between BM25 retrieval and dense retrieval?

LLAMA 3.2:3B
-----
BM25 (Batched Magnetic 25) retrieval and Dense Retrieval are two popular techniques used in Information Retrieval (IR) for ranking se
arch results.

**BM25 Retrieval:**
BM25 is a simple, widely used, and effective retrieval method developed by Robert Morris, Geoffrey Chambers, and Peter G. Russell in
1998. It's an extension of the original BM model, which takes into account the importance of the document's importance scores, length
, and query term frequency.

The main idea behind BM25 is to:

1. Calculate a score for each document based on its importance, length, and term frequency.
2. Rank documents by their scores, assuming the best answer is the top-ranked document.

The formula for BM25 is:

`BM25 = (IDF(q) \* TF(q, d) \* (k1 \* (1 + log(DF(d))) / (k2 + log(DF(d))))`

Where:

* `q` is the query
* `d` is the document

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* `TF(q, d)` is the term frequency of query term `q` in document `d`
* `IDF(q)` is the inverse document frequency of query term `q`
* `k1` and `k2` are hyperparameters
* `DF(d)` is the document frequency of document `d`

**Dense Retrieval:**
Dense Retrieval, on the other hand, is a more recent approach introduced in 2020 by Zhang et al. in the paper "Dense Retrieval for Im
age Search." It represents the similarity between a query and a document as a dense vector, rather than a simple score.

In Dense Retrieval, the goal is to learn a mapping between the query and document spaces, such that the similarity between them can b
e computed using a dot product.

The key idea behind Dense Retrieval is to use a Siamese network architecture, which consists of two identical neural networks:

1. A query network that takes the query vector as input and produces a query embedding.
2. A document network that takes the document vector as input and produces a document embedding.

The similarity between the query and document embeddings is computed using a dot product, and the final ranking is determined by the
similarity score.

**Key differences:**

* BM25 is a score-based retrieval method, while Dense Retrieval is a vector-based retrieval method.
* BM25 relies on simple mathematical calculations, while Dense Retrieval uses neural networks to learn the mapping between the query
and document spaces.

In summary, BM25 is a widely used, simple, and effective retrieval method, while Dense Retrieval is a more recent, vector-based appro
ach that offers improved performance and flexibility.
Response time: 79.54 seconds

QWEN 2.5:3B
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Let's break down the differences between BM25 retrieval and dense retrieval in simpler terms:

1. BM25 Retrieval:
- BM25 stands for "Best Matching 25" and is a method used in information retrieval.
- It helps computers find documents that are most relevant to a specific question or search query.

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- It works by looking at how often certain words (we call these "terms") appear in documents and how similar a query is to those documents.
- It's especially useful when the documents are text-based and we want to understand how well a query matches the content of documents.

2. Dense Retrieval:
- Dense retrieval is about finding similar content (like images, videos, or documents) to a query without using traditional text-based methods.
- It focuses more on the overall similarity between items based on their content, not just the words used.
- For example, if you search for "cat," dense retrieval might find not only text documents about cats but also images of cats or videos of cats.
- It's useful in scenarios where text is not the main way to describe the content, such as when dealing with images or videos.

In summary, BM25 is more about text-based matching and finding documents that contain the most relevant text, while dense retrieval is about finding similar content regardless of how it's described, whether it's text, images, or videos.
Response time: 39.56 seconds

Comparison:
Both models were tested with the same question and the same custom system prompt.

Question 3
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Why might an organization choose a local LLM instead of a cloud API?

LLAMA 3.2:3B
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An organization might choose a local Large Language Model (LLM) over a cloud API for several reasons:

1. Data Privacy and Security: By hosting their own LLM on-premises or locally, organizations can better control and protect their sensitive data. This is especially important for companies dealing with confidential or regulated information.
2. Reduced Dependence on Internet: A local LLM doesn't require a stable internet connection to operate, which can be beneficial for organizations with slow or unreliable internet connections, or those operating in areas with limited internet access.
3. Improved Performance and Latency: By hosting the LLM locally, organizations can reduce latency and improve performance, as data doesn't need to be transferred over the internet.
4. Increased Control and Customization: With a local LLM, organizations have more control over the model's architecture, training data, and updates, allowing them to tailor the model to their specific needs.
5. Reduced Costs: While the initial setup cost of a local LLM might be higher, it can be more cost-effective in the long run, as organizations don't need to pay for cloud API calls or storage.

6. Compliance and Regulatory Requirements: Some industries, such as healthcare or finance, have strict data privacy and security regulations. A local LLM can help organizations comply with these regulations by ensuring data is stored and processed locally.

However, it's worth noting that a local LLM also requires more resources (e.g., hardware, maintenance) compared to a cloud API, which can be managed by the provider.
Response time: 44.09 seconds

QWEN 2.5:3B
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An organization might choose to use a local Large Language Model (LLM) instead of a cloud API for a few reasons, especially if they want more control and security over their data. Here are some reasons explained simply:

1. Data Privacy: If the organization has very sensitive data that they don't want to send to someone else's servers, using a local LLM can be safer. The data stays within their control.
2. Performance: If the organization needs very fast responses or doesn't want any delay from internet traffic, a local model can be quicker.
3. Customization: Some organizations might want to customize the model, add their own data to it, or integrate it more deeply with their systems. This isn't always possible with cloud APIs.
4. Scalability: Local models can scale independently of the internet. They don't slow down due to network traffic or other users using the same model.
5. Cost: If the organization has a lot of data or performs a lot of queries, the cost of transferring that data over the internet can be high. Using a local model can reduce these costs.

However, cloud APIs often come with more features and can be more cost-effective for smaller projects. It's important to weigh these factors and choose the option that best fits the organization's needs.
Response time: 36.02 seconds

Comparison:
Both models were tested with the same question and the same custom system prompt.

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